HW8: Actor Critic Methods

Question 1:

Unbiased Proof:

Prove that Adding Baseline to REINFORCE is unbiased and variance is lower:

(no baseline)

$$\nabla J(\theta) \propto \sum_s \mu(s) \sum_a q_\pi(s,a) \nabla_\theta \pi(a|s,\theta)$$

(baseline)

$$\nabla J(\theta) \propto \sum_s \mu(s) \sum_a \left(q_\pi(s,a) - b(s)\right) \nabla \pi(a|s,\theta)$$

subtracted quantity:

$$\sum_a b(s) \nabla_\theta \pi(a|s,\theta) = b(s) \nabla_\theta \overbrace{\sum_a \pi_\theta(a|s,\theta)}^{=1}$$

all probabilities must sum to 1

$$= b(s) \nabla_\theta 1$$

$$= b(s) \cdot 0$$

$$\sum_a b(s) \nabla_\theta \pi(a|s,\theta) = 0$$

Since subtracted quantity $= 0$, the baseline and non-baseline terms are equivalent in expectation. This means they have the same average and are both unbiased.

Lower Variance Proof:

baseline: $V_\pi(S) \leftarrow$ value function

$$\nabla J(\theta) \propto \sum_s \mu(s) \sum_a \left(q_\pi(s,a) - V_\pi(s)\right) \nabla \pi(a|s,\theta)$$

$$Var[X] = E[X^2] + (E[X])^2$$

$$Var[X-C] = E[X^2 - 2cx + c^2] + (E[X-C])^2$$

$$= E[X^2] - 2E[Xc] + E[c^2] + (E[x] - E[c])^2$$

$$= E[X^2] - 2E[Xc] + E[c^2] + E[X]^2 - 2E[x]E[c] + E[c]^2$$

$$= \underbrace{E[X^2] + (E[X])^2}_{Var[X]} + \underbrace{E[C^2] + (E[C])^2}_{Var[C]} - \underbrace{2\left(E[Xc] - E[X]E[c]\right)}_{Cov(X,c)}$$

Since C is constant w.r.t. action $Var[C] = 0$

Since $q_\pi(s,a)$ and $V_\pi(S)$ both depend on state, there is a positive correlation $\therefore Cov(X,c) > 0$

Since $Cov(x,c) > 0$
$Var[X] > Var[X-C]$

Therefore, adding a baseline to the gradient calculation keeps the expected value unbiased and results in lower variance.

Question 2:

Generalized On-Policy Distribution

$$\eta_\gamma(s) = h(s) + \gamma \sum_{\bar{s}} \eta(\bar{s}) \sum_a \pi(a|\bar{s}) p(s|\bar{s},a)$$

$$\mu_\gamma(s) = \frac{h(s) + \gamma \sum_{\bar{s}} \eta(\bar{s}) \sum_a \pi(a|\bar{s}) p(s|\bar{s},a)}{\sum_{s'} h(s') + \gamma \sum_{\bar{s}} \eta(\bar{s}) \sum_a \pi(a|\bar{s}) p(s'|\bar{s},a)} \qquad \mu_\gamma(s) = \frac{\eta(s)}{\sum_{s'} \eta(s')}$$

$$\nabla v_\pi(s) = \nabla\left[\sum_a \pi(a|s) q_\pi(s,a)\right]$$

$$= \sum_a \left[\nabla\pi(a|s) q_\pi(s,a) + \pi(a|s) \nabla q_\pi(s,a)\right]$$

$$= \sum_a \left[\nabla\pi(a|s) q_\pi(s,a) + \pi(a|s) \nabla \sum_{s',r} p(s',r|s,a)(r + \gamma v_\pi(s'))\right]$$

$$\nabla_\theta \sum_{s',r} \cancel{p(s',r|s,a)r} + p(s',r|s,a)\gamma v_{\pi_\theta}(s'))$$

$$\nabla v_\pi(s) = \sum_a \left[\nabla\pi(a|s) q_\pi(s,a) + \pi(a|s) \gamma \sum_{s'} p(s'|s,a) \nabla v_\pi(s')\right]$$

$$\nabla v_\pi(s) = \sum_a \left[\nabla\pi(a|s) q_\pi(s,a) + \pi(a|s) \gamma \sum_{s'} p(s'|s,a) \cdot \right.$$

$$\sum_{a'} \left[\nabla\pi(a'|s') q_\pi(s',a') + \pi(a'|s') \gamma \sum_{s''} p(s''|s',a') \cdot \nabla v_\pi(s'')\right)$$

$$= \sum_{x \in S} \sum_{k=0}^{\infty} \gamma^k \Pr(s \to x, k, \pi) \sum_a \nabla\pi(a|x) q_\pi(x,a)$$

$$\nabla v_\pi(s_0) = \sum_{x \in S} \sum_{k=0}^{\infty} \gamma^k \Pr(s_0 \to s, k, \pi) \sum_a \nabla \pi(a|x) q_\pi(x,a)$$

$$= \sum_s \eta_\gamma(s) \sum_a \nabla \pi(a|s) q_\pi(s,a)$$

$$= \sum_{s'} \eta_\gamma(s') \sum_s \mu_\gamma(s) \sum_a \nabla \pi(a|s) q_\pi(s,a)$$

$$\propto \sum_s \mu_\gamma(s) \sum_a \nabla \pi(a|s) q_\pi(s,a)$$

back in expectation form

$$\propto \mathbb{E}_\pi\left[\gamma^t \sum_a q_\pi(S_t,a) \nabla \pi(a|S_t,\theta)\right]$$

$$\propto \mathbb{E}_\pi\left[\gamma^t \sum_a \pi(a|S_t,\theta) q_\pi(S_t,a) \frac{\nabla \pi(a|S_t,\theta)}{\pi(a|S_t,\theta)}\right]$$

$$= \mathbb{E}_\pi\left[\gamma^t q_\pi(S_t,A_t) \frac{\nabla \pi(A_t|S_t,\theta)}{\pi(a|S_t,\theta)}\right]$$

$$= \mathbb{E}_\pi\left[\gamma^t G_t \frac{\nabla \pi(A_t|S_t,\theta)}{\pi(A_t|S_t,\theta)}\right]$$

$$\theta_{t+1} = \theta_t + \alpha \gamma^t G_t \frac{\nabla \pi(A_t|S_t,\theta)}{\pi(A_t|S_t,\theta)}$$

Question 3:

Explain why the standard policy gradient cannot be directly estimated when using trajectories sampled from β?

Apply importance sampling to express the policy gradient objective in terms of expectations over trajectories from the behavior policy β to derive off-policy PG.

Standard Policy Gradient:

$$J(\theta) = E_{\tau \sim \pi_\theta}[R(\tau)]$$

$$\nabla J(\theta) = E_{\tau \sim \pi_\theta}[R(\tau) \nabla_\theta \ln \underbrace{P_{\pi_\theta}(\tau)}_{\text{probability of trajectory } \tau \text{ under policy } \pi_\theta}]$$

$$P_{\pi_\theta}(\tau) = \rho(s_0) \prod_{t=0}^{T} \pi_\theta(a_t|s_t) \rho(s_{t+1}|s_t, a_t)$$

If we are instead updating π_θ using experience from π_β this gradient is no longer correct since

$$P_{\pi_\theta}(\tau) \neq P_{\pi_\beta}(\tau)$$

In order to use $\tau \sim \pi_\beta$ to update π_θ, we must do importance sampling:

$$\nabla J(\theta) = E_{\tau \sim \pi_\theta}\left[\underbrace{\frac{P_{\pi_\theta}(\tau)}{P_{\pi_\beta}(\tau)}}_{\text{importance ratio}} R(\tau) \nabla_\theta \ln P_{\pi_\theta}(\tau)\right]$$

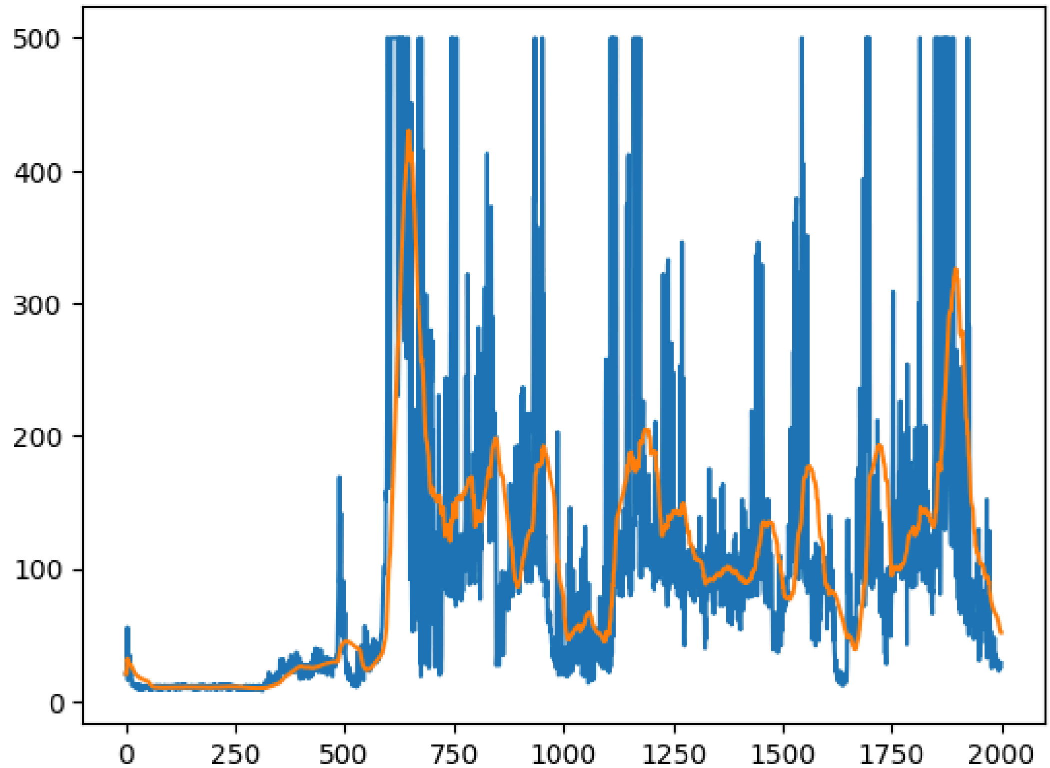
$$\frac{P_{\pi_\theta}(\tau)}{P_{\pi_\beta}(\tau)} = \frac{\cancel{\rho(s_0)} \prod_{t=0}^{T} \pi_\theta(a_t|s_t) \cancel{\rho(s_{t+1}|s_t, a_t)}}{\cancel{\rho(s_0)} \prod_{t=0}^{T} \pi_\beta(a_t|s_t) \cancel{\rho(s_{t+1}|s_t, a_t)}}$$

$$\nabla J(\theta) = E_{\tau \sim \pi_\theta}\left[\prod_{t=0}^{T} \frac{\pi_\theta(a_t|s_t)}{\pi_\beta(a_t|s_t)} R(\tau) \nabla_\theta \ln P_{\pi_\theta}(\tau)\right]$$

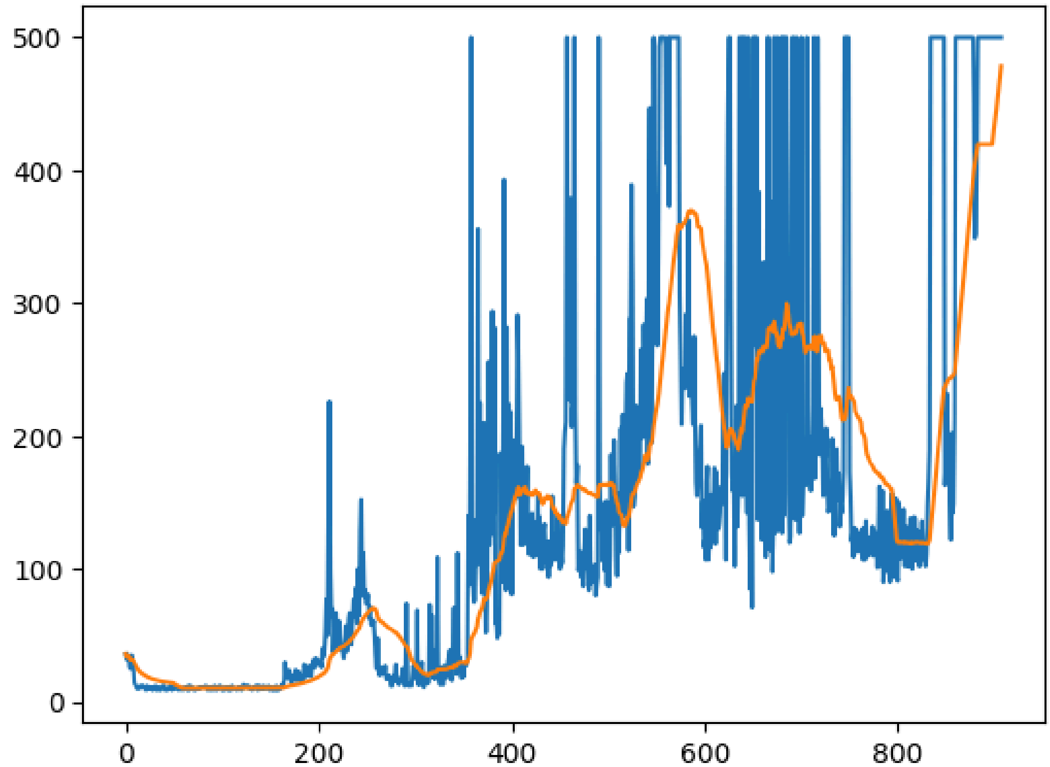
$$\nabla J(\theta) = E_{\tau \sim \pi_\theta}\left[\prod_{t=0}^{T} \frac{\pi_\theta(a_t|s_t)}{\pi_\beta(a_t|s_t)} R(\tau) \nabla_\theta \ln \pi_\theta(a_t|s_t)\right]$$

Question 4: Cart-Pole with Policy Gradient

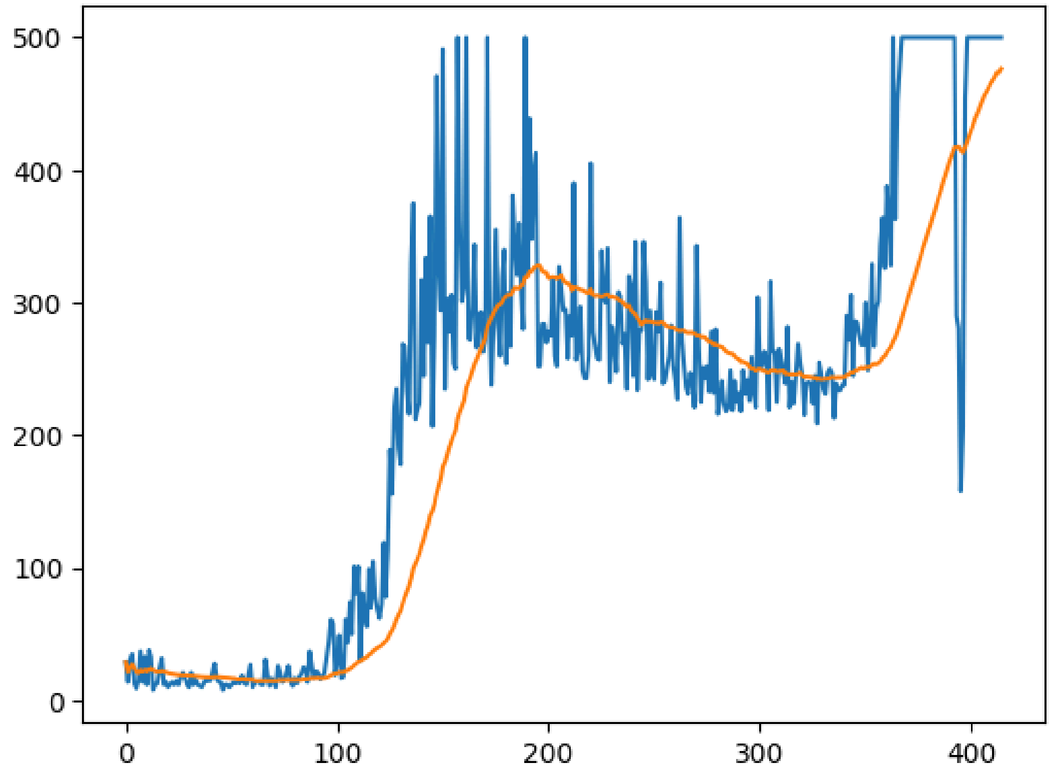
DDPG:



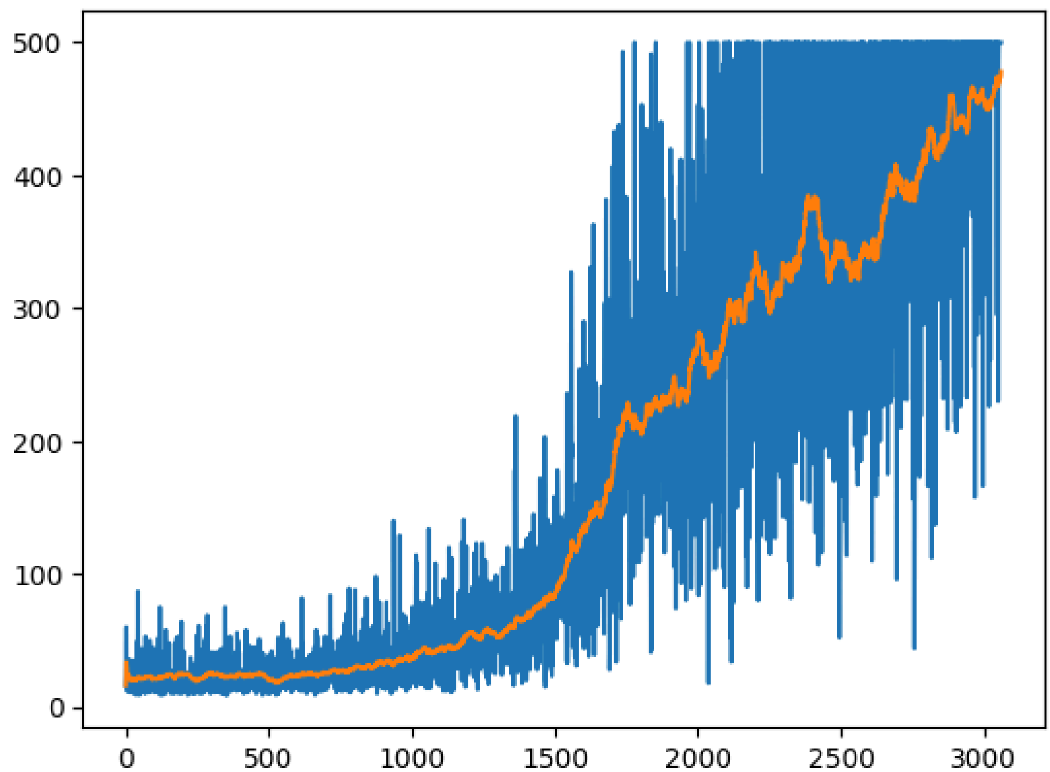
TD3:



SAC:



PPO:



Question 5: Eligibility Vector for Softmax Policy

$$\pi(a|s,\theta) = \frac{e^{h(s,a,\theta)}}{\sum_b e^{h(s,b,\theta)}}$$

$$h(s,a,\theta) = \theta^T x(s,a) \qquad x(s,a) \in \mathbb{R}^d \text{ (feature vector)}$$

$$\nabla_\theta \ln \pi(a|s,\theta) = \nabla_\theta \ln\left[\frac{e^{\theta^T x(s,a)}}{\sum_b e^{\theta^T x(s,b)}}\right]$$

$$= \nabla_\theta\left[\ln e^{\theta^T x(s,a)}\right] - \nabla_\theta\left[\ln \sum_b e^{\theta^T x(s,b)}\right]$$

$$= \nabla_\theta\, \theta^T x(s,a) - \frac{\nabla_\theta \sum_b e^{\theta^T x(s,b)}}{\sum_b e^{\theta^T x(s,b)}}$$

$$= x(s,a) - \frac{\sum_b e^{\theta^T x(s,b)} \cdot x(s,b)}{\sum_b e^{\theta^T x(s,b)}}$$

$\rightarrow \pi(a|s,\theta)$

$$= x(s,a) - \sum_b \frac{e^{\theta^T x(s,b)}}{\sum_b e^{\theta^T x(s,b)}} \cdot x(s,b)$$

$$\boxed{\nabla_\theta \ln \pi(a|s,\theta) = x(s,a) - \sum_b \pi(b|s,\theta) \cdot x(s,b)}$$